Problem 1. Use the ε, δ definition of the limit to check that

$$\lim_{x \to 1}(2x^2 + 3) = 5$$

via the following steps. This will be similar to the continuity of $f(x) = x^2$ in the lecture notes.

(a) Write down the ε, δ definition of the limit in this case.

(b) Factor $|(2x^2 + 3) - 5|$ as $|x - 1| \cdot *$ for some expression $*$.

(c) Show that if $|x - 1| < 1$, then $* \leq C$ for some fixed number C.[1] (You should find a specific number here.)

(d) Show that if $|x - 1| < \min(1, \varepsilon/C)$, then $|(2x^2 + 3) - 5| < \varepsilon$.

Solution. (a) For any $\varepsilon > 0$, there is a $\delta > 0$ such that if $0 < |x - 1| < \delta$, $|(2x^2 + 3) - 5| < \varepsilon$.

(b) First of all, $|(2x^2+3)-5| = |2x^2-2| = 2|x^2-1|$. We can then use the factorization $x^2-1 = (x+1)(x-1)$ to write this as $|x - 1| \cdot 2|x + 1|$.

(c) There are two approaches we could use here. One is to use the triangle inequality as follows: assuming $|x - 1| < 1$,

$$|x + 1| = |(x - 1) + 2| \leq |x - 1| + 2 < 3$$

and so $2|x + 1| \leq 6$. The other approach is to note that if $|x - 1| < 1$, then x is in the interval $(0, 2)$, and so $x + 1$ is in the interval $(1, 3)$. Therefore, $2|x + 1| < 2 \cdot 3 = 6$. In any case, we take $C = 6$.

(d) If $|x - 1| < \min(1, \varepsilon/6)$, then in particular $|x - 1| < 1$, and so from part (c) we have $2|x + 1| \leq 6$. We also have $|x - 1| < \varepsilon/6$, and so we have the following chain of inequalities:

$$|(2x^2 + 3) - 5| = |x - 1| \cdot 2|x + 1| \leq |x - 1| \cdot 6 < (\varepsilon/6) \cdot 6 = \varepsilon$$

Therefore, $|(2x + 3) - 5| < \varepsilon$, as required.

Commentary. In general, for any function f, we can write

$$|f(x) - f(a)| = |x - a| \cdot \left| \frac{f(x) - f(a)}{x - a} \right|$$

The second term is the average rate of change of f between x and a. As long as we have some bound on the average rate of change for x close to a, then we can use an argument as in this problem to show that the function is continuous at a. In particular, if the function is differentiable at a, then for x near a, the average rate of change is close to $f'(a)$, and so we can get a bound. This is essentially why differentiability implies continuity.

[1] You can use the *triangle inequality* here. This says that for any numbers a, b, $|a + b| \leq |a| + |b|$.